

Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить перенаправление ввода-вывода
- 2 Изучить работу фильтров
- 3 Изучить команду поиска
- 4 Ознакомиться с управлением процессами
- 5 Ознакомиться с командами `df` `du`

2. Процесс выполнения лабораторной работы

```
asudenur@asudenur:~$ ls /etc/ > file.txt
asudenur@asudenur:~$ ls >> file.txt
asudenur@asudenur:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
ceph
chromium
chrony.conf
cifs-utils
cockpit
colord
```

Рисунок 1: Запись в файл

```
asudenur@asudenur:~$  
asudenur@asudenur:~$ grep .conf file.txt > conf.txt  
asudenur@asudenur:~$ cat conf.txt  
anthy-unicode.conf  
asound.conf  
brltty.conf  
chrony.conf  
cpupower-service.conf  
dconf  
dnsmasq.conf  
dracut.conf  
dracut.conf.d  
fprintd.conf  
fuse.conf  
host.conf  
idmapd.conf  
kdump.conf  
krb5.conf  
krb5.conf.d  
ld.so.conf  
ld.so.conf.d  
libaudit.conf  
libuser.conf  
locale.conf  
logrotate.conf  
makedumpfile.conf.sample  
man_db.conf
```

Рисунок 2: Поиск расширения .conf


```
/home/asudenur/.npm/_cacache/index-v5/f3/6c/c3986891a2eed646eaa7d914d91cffc8d6421272c959bdb5a
87486645945
/home/asudenur/.npm/_cacache/index-v5/ce
/home/asudenur/.npm/_cacache/index-v5/84/ce
/home/asudenur/.npm/_cacache/index-v5/09/c0
/home/asudenur/.npm/_cacache/index-v5/09/c0/cce94e2c664a8f390a8f1ade3346d5c95f3ced5cf6fc29def
30220242c53
/home/asudenur/.npm/_cacache/index-v5/59/e7/cb225f160e685933e8439d3cf624ef721a58996cfad55fb20
00f59e2edcc
/home/asudenur/.npm/_cacache/index-v5/cd
/home/asudenur/.npm/_cacache/index-v5/c9
/home/asudenur/.npm/_cacache/index-v5/c0
/home/asudenur/.npm/_cacache/index-v5/9f/ae/c077854801f7b4b5cf7eea2bb68ca2bc53f0b53cea310cda0
b203217f693
/home/asudenur/.npm/_cacache/index-v5/44/44/c9a2c5dc2999cbe868c370ef479e17dfd1e86117d1350482a
1d990504757
/home/asudenur/.npm/_cacache/index-v5/af/cf
/home/asudenur/.npm/_cacache/index-v5/c3
/home/asudenur/.npm/_cacache/index-v5/78/34/c1e131014ad3a4ee13f096e70f74c03e68837bc2c8188a44d
554f3f3ba2e
/home/asudenur/.npm/_cacache/index-v5/a1/5e/c9aa1c46bf0c67511379e97bd07fca87532422f2506214f94
db8861160d3
/home/asudenur/.npm/_cacache/index-v5/ca
/home/asudenur/snap/hugo/common
/home/asudenur/snap/hugo/current
/home/asudenur/conf.txt
asudenur@asudenur:~$
```

Рисунок 3: Поиск файлов

```
/etc/foomatic/hashes.d/hashes.new
find: '/etc/libvirt': Permission denied
/etc/hp
/etc/hp/hpclip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
find: '/etc/lvm/archive': Permission denied
/etc/logrotate.d/httpd
find: '/etc/lvm/backup': Permission denied
find: '/etc/lvm/cache': Permission denied
find: '/etc/lvm/devices': Permission denied
find: '/etc/nftables': Permission denied
find: '/etc/openvpn/client': Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: '/etc/openvpn/server': Permission denied
find: '/etc/pki/rsyslog': Permission denied
find: '/etc/polkit-1/localauthority': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/sos/cleaner': Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
/etc/sane.d/hpsj5s.conf
/etc/sane.d/hs2p.conf
:find: '/etc/ssh/sshd_config.d': Permission denied
find: '/etc/sss': Permission denied
find: '/etc/sudoers.d': Permission denied
find: '/etc/userdb': Permission denied
```

Рисунок 4: Поиск файлов

```
asudenur@asudenur:~$  
asudenur@asudenur:~$ find /etc -name "h*" -print | less  
find: '/etc/audit': Permission denied  
asudenur@asudenur:~$  
asudenur@asudenur:~$  
asudenur@asudenur:~$ find ~ -name "log*" > logfile &  
[1] 6760  
asudenur@asudenur:~$  
[1]+  Done                  find ~ -name "log*" > logfile  
asudenur@asudenur:~$ rm logfile  
asudenur@asudenur:~$
```

Рисунок 5: Фоновый запуск процесса

```
asudenur@asudenur:~$  
asudenur@asudenur:~$ gedit &  
[1] 6803  
asudenur@asudenur:~$ ps | grep gedit  
  6803 pts/1    00:00:00 gedit  
asudenur@asudenur:~$ kill 6803  
[1]+  Terminated                  gedit  
asudenur@asudenur:~$  
asudenur@asudenur:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

```
DF(1)                                User Commands                                DF(1)

NAME
    df - report file system space usage

SYNOPSIS
    df [OPTION]... [FILE]...

DESCRIPTION
    This manual page documents the GNU version of df.  df displays the amount of space
    available on the file system containing each file name argument.  If no file name is
    given, the space available on all currently mounted file systems is shown.  Space is
    shown in 1K blocks by default, unless the environment variable POSIXLY_CORRECT is
    set, in which case 512-byte blocks are used.

    If an argument is the absolute file name of a device node containing a mounted file
    system, df shows the space available on that file system rather than on the file sys-
    tem containing the device node.  This version of df cannot show the space available
    on unmounted file systems, because on most kinds of systems doing so requires non-
    portable intimate knowledge of file system structures.

OPTIONS
    Show information about the file system on which each FILE resides, or all file sys-
    tems by default.

    Mandatory arguments to long options are mandatory for short options too.

    -a, --all
        include pseudo, duplicate, inaccessible file systems

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of
        1,048,576 bytes; see SIZE format below

Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

```
du(1)                                User Commands                                du(1)

NAME
    du - estimate file space usage

SYNOPSIS
    du [OPTION]... [FILE]...
    du [OPTION]... --files0-from=F

DESCRIPTION
    Summarize device usage of the set of FILES, recursively for directories.

    Mandatory arguments to long options are mandatory for short options too.

    -0, --null
        end each output line with NUL, not newline

    -a, --all
        write counts for all files, not just directories

    --apparent-size
        print apparent sizes rather than device usage; although the apparent size is
        usually smaller, it may be larger due to holes in ('sparse') files, internal
        fragmentation, indirect blocks, and the like

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of
        1,048,576 bytes; see SIZE format below

    -b, --bytes
        equivalent to '--apparent-size --block-size=1'

    -c, --total
        Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

```
asudenur@asudenur:~$ man du
asudenur@asudenur:~$ df
Filesystem      1K-blocks    Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 91233948 63349892 60% /
devtmpfs        4011288      0    4011288  0% /dev
tmpfs           4035660      100   4035560  1% /dev/shm
tmpfs           1614268     1960   1612308  1% /run
tmpfs           1024         0       1024  0% /run/credentials/systemd-journald.service
tmpfs           4035664      80   4035584  1% /tmp
/dev/nvme0n1p3 155186176 91233948 63349892 60% /home
/dev/loop2      49280      49280         0 100% /var/lib/snapd/snap/snapd/25935
/dev/nvme0n1p2 1992552    518800   1352512 28% /boot
/dev/loop0      68480      68480         0 100% /var/lib/snapd/snap/core24/1349
/dev/loop1     115712    115712         0 100% /var/lib/snapd/snap/hugo/25676
tmpfs           1024         0       1024  0% /run/credentials/systemd-resolved.service
tmpfs           807132      180   806952  1% /run/user/1061
/dev/sr0        2677920    2677920         0 100% /run/media/asudenur/Fedora-WS-Live-43
asudenur@asudenur:~$
```

Рисунок 9: Справка по команде du

```
4      ./npm/_cacache/index-v5/a7/81
4      ./npm/_cacache/index-v5/a7
4      ./npm/_cacache/index-v5/30/1a
4      ./npm/_cacache/index-v5/30
424    ./npm/_cacache/index-v5
115644 ./npm/_cacache
115708 ./npm
16     ./snap/hugo/25676/.config/go/telemetry/local
0      ./snap/hugo/25676/.config/go/telemetry/upload
16     ./snap/hugo/25676/.config/go/telemetry
16     ./snap/hugo/25676/.config/go
16     ./snap/hugo/25676/.config
16     ./snap/hugo/25676
0      ./snap/hugo/common
20     ./snap/hugo
20     ./snap
0      ./monthly
0      ./reports/monthly/monthly
0      ./reports/monthly
0      ./reports
4      ./ski.plases/equipment
0      ./ski.plases/plans
4      ./ski.plases
0      ./australia
0      ./play/games/play
0      ./play/games
0      ./play
1467816 .
asudenur@asudenur:~$
```

Рисунок 10: Запуск команды du


```
/home/asudenur/.npm/_cacache/index-v5/d4/09
/home/asudenur/.npm/_cacache/index-v5/4b
/home/asudenur/.npm/_cacache/index-v5/4b/ef
/home/asudenur/.npm/_cacache/index-v5/85
/home/asudenur/.npm/_cacache/index-v5/85/e2
/home/asudenur/.npm/_cacache/index-v5/a7
/home/asudenur/.npm/_cacache/index-v5/a7/81
/home/asudenur/.npm/_cacache/index-v5/30
/home/asudenur/.npm/_cacache/index-v5/30/1a
/home/asudenur/snap
/home/asudenur/snap/hugo
/home/asudenur/snap/hugo/25676
/home/asudenur/snap/hugo/25676/.config
/home/asudenur/snap/hugo/25676/.config/go
/home/asudenur/snap/hugo/25676/.config/go/telemetry
/home/asudenur/snap/hugo/25676/.config/go/telemetry/local
/home/asudenur/snap/hugo/25676/.config/go/telemetry/upload
/home/asudenur/snap/hugo/common
/home/asudenur/monthly
/home/asudenur/reports
/home/asudenur/reports/monthly
/home/asudenur/reports/monthly/monthly
/home/asudenur/ski.places
/home/asudenur/ski.places/equipment
/home/asudenur/ski.places/plans
/home/asudenur/australia
/home/asudenur/play
/home/asudenur/play/games
/home/asudenur/play/games/play
asudenur@asudenur:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.